D8 Transactions Implementation

Track B | MySQL 8.4.9 / InnoDB

The cycle-close and bid-close code is in **d8_transactions.sql**. Apply it once to the D7 v2 schema. It adds the required columns and installs the procedures. Each close uses one transaction.

Rollback demonstration log

1. Pot not fully funded

Cycle 11 was still missing two confirmed payments after the final receipt was entered.

```
ERROR 1644 (45000) at line 2: D8_POT_NOT_FUNDED
```

Recorded state	Before	After
Cycle status	open	open
Contribution receipts	1=paid, 2=unpaid, 3=unknown; no receipt for member 4	1=paid, 2=unpaid, 3=unknown; no receipt for member 4
Paid pot	100.00	100.00
Winning bid	None	None
Payout recipient	None	None
Discount shares	None	None
Eligible members	1, 2, 3, 4	1, 2, 3, 4

2. Duplicate payout

Cycle 22 already had a payout for member 3. The new insert hit the payout primary key.

```
ERROR 1062 (23000) at line 2: Duplicate entry '22' for key 'payout.PRIMARY'
```

Recorded state	Before	After
Cycle status	open	open
Contribution receipts	1=paid, 2=paid, 3=paid; no receipt for member 4	1=paid, 2=paid, 3=paid; no receipt for member 4
Paid pot	300.00	300.00
Winning bid	None	None
Payout recipient	3	3
Discount shares	None	None
Eligible members	2, 3, 4	2, 3, 4

Isolation level

I used **READ COMMITTED** with **SELECT ... FOR UPDATE**. The close locks the circle first and then the cycle. Payment, bid and close operations take the same circle lock, so they cannot change the funding or bids while the close is running. A second close waits, then reads the committed cycle status and stops because the cycle is already closed.

Track B concurrency test

The Node.js script ran two independent MySQL sessions against the same cycle on a disposable copy of the D7 schema. A third session held the circle lock until MySQL reported that both requests were waiting. Releasing the lock let the two requests compete.

The test script is submitted separately as **test_d8.js**. In each of the 20 trials, one request committed and the other returned **D8_CYCLE_NOT_OPEN**.

Results

Check	Observed result
Bidding closed	All 20 cycles changed from open to closed.
Winning bid selected	The expected bid was recorded in every run. For cycle 102, it was bid 104.
Discount redistributed	Three shares per cycle: 16.46, 16.45 and 16.45. Total: 49.36.
Payout recorded	Exactly one payout per cycle, to member 2. Net payout was 350.64.
Eligibility updated	The winner went from 0 to 1 payout turn and became ineligible. Other members kept their prior counts.

The final receipt was recorded in each successful close. Four paid contributions funded the 400.00 pot.

Sample terminal output

```
cycle=102 waiters=2 outcomes=fulfilled,rejected
ERROR 1644 (45000) at line 2: D8_CYCLE_NOT_OPEN
PASS: 20 commits, 20 rejections, 20/20 state checks.
```